The string will continue to slide around and oscillate, but it will do so in this extended configuration. In fact, the string can wrap around the circular part of the space any number of times, as also shown in Figure 10.3(b), and again will execute oscillatory motion as it slides around. When a string is in such a wrapped configuration, we say that it is in a winding mode of motion. Clearly, being in a winding mode is a possibility inherent to strings. There is no point-particle counterpart. We now seek to understand the implications of this qualitatively new kind of string motion on the string itself as well as on the geometrical properties of the dimension it wraps.

Throughout our previous discussion of string motion, we have focused on unwound strings. Strings that wrap around a circular component of space share almost all of the same properties as the strings we have studied. Their oscillations, just as those of their unwound counterparts, contribute strongly to their observed properties. The essential difference is that a wrapped string has a minimum mass, determined by the size of the circular dimension and the number of times it wraps around. The string's oscillatory motion determines a contribution in excess of this minimum.

It is not difficult to understand the origin of this minimum mass. A wound string has a minimum length determined by the circumference of the circular dimension and the number of times the string encircles it. The minimum length of a string determines its minimum mass: The longer this length, the greater the mass, since there is more of it. Since the circumference of a circle is proportional to its radius, the minimum winding-mode masses are proportional to the radius of the circle being wrapped. By using Einstein's $E = mc$

2 relating mass to energy, we can also say that the energy bound in a wound string is proportional to the radius of the circular dimension. (Unwrapped strings also have a tiny minimum length since if they didn't, we would be back in the realm of point particles. The same reasoning might lead to the conclusion that even unwrapped strings have a minuscule yet nonzero minimum mass. In a sense this is true, but the quantum-mechanical effects encountered in Chapter 6—remember The Price Is Right, again—are able to exactly cancel this contribution to the mass. This is how, we recall, unwrapped strings can yield the zero-mass photon, graviton, and the other massless or near-